KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5121 | 5 Credits: 4 | Program Core | Source-grounded

Adhesive Technology

 To provide knowledge about the application of various adhesives in wood based

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5121 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5121.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Wood and Paper Technology
Course code	5121
Course title	Adhesive Technology
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

14 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide knowledge about the application of various adhesives in wood based
- To impart knowledge about characteristics of glues and method of application on wood and wood products
- To familiarize the students manufacture of Adhesives

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding composition	See official syllabus
CO2	Discuss the physical chemical principles of gluing 17 Applying Describe the manufacture and industrial	See official syllabus
CO3	17 Applying applications of Thermosetting resins Summarize the manufacture and industrial	See official syllabus
CO4	9 Understanding applications of Thermoplastic resins	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Classify the types of adhesives and explain their composition	4	As specified
Module 2	Discuss the physical chemical principles of gluing	3	As specified
Module 3	resins	4	As specified
Module 4	resins	3	As specified

MODULE 1

Classify the types of adhesives and explain their composition

This module is presented from the official syllabus scope for Adhesive Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Classify the types of adhesives and explain their composition

- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

1 industry Understanding

1 industry Understanding is an official topic in Module 1 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 industry Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 industry understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 industry Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 1 industry Understanding [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Classify adhesives 6 Understanding Explain the composition of adhesives and additives

Classify adhesives 6 Understanding Explain the composition of adhesives and additives is an official topic in Module 1 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify adhesives 6 Understanding Explain the composition of adhesives and additives using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify adhesives 6 understanding explain the composition of adhesives and additives should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classify adhesives 6 Understanding Explain the composition of adhesives and additives means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Classify adhesives 6 Understanding Explain the composition of adhesives and additives □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Applying used

4 Applying used is an official topic in Module 1 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying used using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying used should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying used means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 4 Applying used [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Discuss the concepts of glue mixing 4 Applying Adhesive types and their composition Adhesive industries in India - an overview Adhesion, adhesives, adherend, glue line; Classification of adhesives - Natural- starches, dextrins, vegetable gums, protein glues- animal glues, blood albumin, casein, vegetable protein glue, shellac rubber Asphalt sodium silicate, magnesium oxychloride; Synthetic – Thermosetting (urea, melamine,phenol, resorcinol..) and Thermoplastic resin(cellulose esters and ethers, polyvinyl alcohol..) Composition of adhesive - back bone polymer, solvent, thinner, catalyst, hardner, extender, filler, plasticiser, fortifier, preservatives; Glue mixing- types of mixers - process by which adhesive develop strength, addition of caustic soda or acid to control the PH during mixing. Viscosity, shelf life and pot life of resin.

Discuss the concepts of glue mixing 4 Applying Adhesive types and their composition Adhesive industries in India - an overview Adhesion, adhesives, adherend, glue line; Classification of adhesives - Natural- starches, dextrins, vegetable gums, protein glues- animal glues, blood albumin, casein, vegetable protein glue, shellac rubber Asphalt sodium silicate, magnesium oxychloride; Synthetic – Thermosetting (urea, melamine,phenol, resorcinol..) and Thermoplastic resin(cellulose esters and ethers, polyvinyl alcohol..) Composition of adhesive - back bone polymer, solvent, thinner, catalyst, hardner, extender, filler, plasticiser, fortifier, preservatives; Glue mixing- types of mixers - process by which adhesive develop strength, addition of caustic soda or acid to control the PH during mixing. Viscosity, shelf life and pot life of resin. is an official topic in Module 1 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the concepts of glue mixing 4 Applying Adhesive types and their composition Adhesive industries in India - an overview Adhesion, adhesives, adherend, glue line; Classification of adhesives - Natural- starches, dextrins, vegetable gums, protein glues- animal glues, blood albumin, casein, vegetable protein glue, shellac rubber Asphalt sodium silicate, magnesium oxychloride; Synthetic – Thermosetting (urea, melamine,phenol, resorcinol..) and

Thermoplastic resin(cellulose esters and ethers, polyvinyl alcohol..) Composition of adhesive - back bone polymer, solvent, thinner, catalyst, hardner, extender, filler, plasticiser, fortifier, preservatives; Glue mixing- types of mixers - process by which adhesive develop strength, addition of caustic soda or acid to control the PH during mixing. Viscosity, shelf life and pot life of resin. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the concepts of glue mixing 4 applying adhesive types and their composition adhesive industries in india - an overview adhesion, adhesives, adherend, glue line; classification of adhesives - natural- starches, dextrins, vegetable gums, protein glues- animal glues, blood albumin, casein, vegetable protein glue, shellac rubber asphalt sodium silicate, magnesium oxychloride; synthetic – thermosetting (urea, melamine,phenol, resorcinol..) and thermoplastic resin(cellulose esters and ethers, polyvinyl alcohol..) composition of adhesive - back bone polymer, solvent, thinner, catalyst, hardner, extender, filler, plasticiser, fortifier, preservatives; glue mixing- types of mixers - process by which adhesive develop strength, addition of caustic soda or acid to control the ph during mixing. viscosity, shelf life and pot life of resin. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss the concepts of glue mixing 4 Applying Adhesive types and their composition Adhesive industries in India - an overview Adhesion, adhesives, adherend, glue line; Classification of adhesives - Natural- starches, dextrins, vegetable gums, protein glues- animal glues, blood albumin, casein, vegetable protein glue, shellac rubber Asphalt sodium silicate, magnesium oxychloride; Synthetic – Thermosetting (urea, melamine,phenol, resorcinol..) and Thermoplastic resin(cellulose esters and ethers, polyvinyl alcohol..) Composition of adhesive - back bone polymer, solvent, thinner, catalyst, hardner, extender, filler, plasticiser, fortifier, preservatives; Glue mixing- types of mixers - process by which adhesive develop strength, addition of caustic soda or acid to control the PH during mixing. Viscosity, shelf life and pot life of resin. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Discuss the concepts of glue mixing 4 Applying Adhesive types and their composition Adhesive industries in India - an overview Adhesion, adhesives, adherend, glue line; Classification of adhesives - Natural-starches, dextrins, vegetable gums, protein glues- animal glues, blood albumin, casein, vegetable protein glue, shellac rubber Asphalt sodium silicate, magnesium oxychloride; Synthetic – Thermosetting (urea, melamine,phenol, resorcinol..) and Thermoplastic resin(cellulose esters and ethers, polyvinyl alcohol..) Composition of adhesive - back bone polymer, solvent, thinner, catalyst, hardner, extender, filler, plasticiser, fortifier, preservatives; Glue mixing- types of mixers - process by which adhesive develop strength, addition of caustic soda or acid to control the PH during mixing. Viscosity, shelf life and pot life of resin. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Classify the types of adhesives and explain their composition
             Adhesives and their importance in wood
  M1.01                                                                  1
             industry
Understanding
  M1.02      Classify adhesives                                          6
Understanding
             Explain the composition of adhesives and additives
  M1.03                                                                  4
Applying
             used
  M1.04      Discuss the concepts of glue mixing                         4
Applying

Contents:
Adhesive types and their composition
Adhesive industries in India - an overview
Adhesion, adhesives, adherend, glue line;
Classification of adhesives -
Natural- starches, dextrins, vegetable gums, protein glues- animal glues, blood
albumin,
casein, vegetable protein glue, shellac rubber Asphalt sodium silicate, magnesium
oxychloride;
Synthetic – Thermosetting (urea, melamine,phenol, resorcinol..) and Thermoplastic
resin(cellulose esters and ethers, polyvinyl alcohol..)
Composition of adhesive - back bone polymer, solvent, thinner, catalyst, hardner,
extender,
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Discuss the physical chemical principles of gluing

This module is presented from the official syllabus scope for Adhesive Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Discuss the physical chemical principles of gluing
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain the rheology of glues

Explain the rheology of glues is an official topic in Module 2 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the rheology of glues using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the rheology of glues should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the rheology of glues means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain the rheology of glues □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Discuss the glue spreading 5 Understanding Demonstrate the tests on glues and gluing

Discuss the glue spreading 5 Understanding Demonstrate the tests on glues and gluing is an official topic in Module 2 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the glue spreading 5 Understanding Demonstrate the tests on glues and gluing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the glue spreading 5 understanding demonstrate the tests on glues and gluing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss the glue spreading 5 Understanding Demonstrate the tests on glues and gluing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Discuss the glue spreading 5 Understanding Demonstrate the tests on glues and gluing [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

6 Applying defects Physical chemical principles of gluing Cohesion - adhesion, Rheology, intermolecular forces, polarity, primary and secondary forces (valences), phenomenon of hardening Influence of - temperature& concentration on surface tension; PH and amount of polymerization on hardening of glue joints, physical properties of adhesive film- thickness of adhesive film Pre-treatments of wood prior to gluing. Glue spreading, Defects in gluing - effect of wood species and moisture content on the strength of glue joint - strength improvement and stresses in glue joints. Tests on adhesive - PH, Viscosity, water tolerance, solid content by oven dry method, specific gravity, solubility, estimation of phenol and formaldehyde. Describe the manufacture and industrial applications of Thermosetting

6 Applying defects Physical chemical principles of gluing Cohesion - adhesion, Rheology, intermolecular forces, polarity, primary and secondary forces (valences), phenomenon of hardening Influence of - temperature& concentration on surface tension; PH and amount of polymerization on hardening of glue joints, physical properties of adhesive film- thickness of adhesive film Pre-treatments of wood prior to gluing. Glue spreading, Defects in gluing - effect of wood species and moisture content on the strength of glue joint - strength improvement and stresses in glue joints. Tests on adhesive - PH, Viscosity, water tolerance, solid content by oven dry method, specific gravity, solubility, estimation of phenol and formaldehyde. Describe the manufacture and industrial applications of Thermosetting is an official topic in Module 2 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Applying defects Physical chemical principles of gluing Cohesion - adhesion, Rheology, intermolecular forces, polarity, primary and secondary forces (valences), phenomenon of hardening Influence of - temperature& concentration on surface tension; PH and amount of polymerization on hardening of glue joints, physical properties of adhesive film- thickness of adhesive film Pre-treatments of wood prior to gluing. Glue spreading, Defects in gluing - effect of wood species and moisture content on the strength of glue joint - strength improvement and stresses in glue joints. Tests on adhesive - PH, Viscosity,

water tolerance, solid content by oven dry method, specific gravity, solubility, estimation of phenol and formaldehyde. Describe the manufacture and industrial applications of Thermosetting using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 applying defects physical chemical principles of gluing cohesion - adhesion, rheology, intermolecular forces, polarity, primary and secondary forces (valences), phenomenon of hardening influence of - temperature& concentration on surface tension; ph and amount of polymerization on hardening of glue joints, physical properties of adhesive film- thickness of adhesive film pre-treatments of wood prior to gluing. glue spreading, defects in gluing - effect of wood species and moisture content on the strength of glue joint - strength improvement and stresses in glue joints. tests on adhesive - ph, viscosity, water tolerance, solid content by oven dry method, specific gravity, solubility, estimation of phenol and formaldehyde. describe the manufacture and industrial applications of thermosetting should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Applying defects Physical chemical principles of gluing Cohesion - adhesion, Rheology, intermolecular forces, polarity, primary and secondary forces (valences), phenomenon of hardening Influence of - temperature& concentration on surface tension; PH and amount of polymerization on hardening of glue joints, physical properties of adhesive film- thickness of adhesive film Pre-treatments of wood prior to gluing. Glue spreading, Defects in gluing - effect of wood species and moisture content on the strength of glue joint - strength improvement and stresses in glue joints. Tests on adhesive - PH, Viscosity, water tolerance, solid content by oven dry method, specific gravity, solubility, estimation of phenol and formaldehyde. Describe the manufacture and industrial applications of Thermosetting means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 6 Applying defects Physical chemical principles of gluing Cohesion - adhesion, Rheology, intermolecular forces, polarity, primary and secondary forces (valences), phenomenon of hardening Influence of - temperature& concentration on surface tension; PH and amount of polymerization on hardening of glue joints, physical properties of adhesive film- thickness of adhesive film Pre-treatments of wood prior to gluing. Glue spreading, Defects in gluing - effect of wood species and moisture content on the strength of glue joint - strength improvement and stresses in glue joints. Tests on adhesive - PH, Viscosity, water tolerance, solid content by oven dry method, specific gravity, solubility, estimation of phenol and formaldehyde. Describe the manufacture and industrial applications of Thermosetting □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Discuss the physical chemical principles of gluing

  M2.01      Explain the rheology of glues                              6
Understanding
  M2.02      Discuss the glue spreading                                5
Understanding
             Demonstrate the tests on glues and gluing
  M2.03                                                                6
Applying
             defects
             Series Test – I                                           1

Contents:
Physical chemical principles of gluing
Cohesion - adhesion, Rheology, intermolecular forces, polarity, primary and
secondary
forces (valences), phenomenon of hardening
 Influence of - temperature& concentration on surface tension; PH and amount of
polymerization on hardening of glue joints, physical properties of adhesive film-
thickness
of adhesive film
Pre-treatments of wood prior to gluing.
Glue spreading, Defects in gluing - effect of wood species and moisture content
on the
strength of glue joint - strength improvement and stresses in glue joints.
Tests on adhesive - PH, Viscosity, water tolerance, solid content by oven dry
method,
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

resins

This module is presented from the official syllabus scope for Adhesive Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: resins
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Classify synthetic resins and its types 2 Understanding Summarize the manufacture of thermosetting

Classify synthetic resins and its types 2 Understanding Summarize the manufacture of thermosetting is an official topic in Module 3 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify synthetic resins and its types 2 Understanding Summarize the manufacture of thermosetting using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify synthetic resins and its types 2 understanding summarize the manufacture of thermosetting should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classify synthetic resins and its types 2 Understanding Summarize the manufacture of thermosetting means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Classify synthetic resins and its types 2 Understanding Summarize the manufacture of thermosetting □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

8 Understanding resins Discuss the applications of thermosetting resins

8 Understanding resins Discuss the applications of thermosetting resins is an official topic in Module 3 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding resins Discuss the applications of thermosetting resins using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding resins discuss the applications of thermosetting resins should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Understanding resins Discuss the applications of thermosetting resins means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 8 Understanding resins Discuss the applications of thermosetting resins □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Applying in wood industry

4 Applying in wood industry is an official topic in Module 3 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying in wood industry using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying in wood industry should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying in wood industry means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 4 Applying in wood industry [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the working of Resin kettle 3 Understanding Synthetic resins - Introduction, chemistry, method of manufacture, lab preparation, properties& industrial applications of: Thermosetting resins Amino resins - Urea formaldehyde (UF) and Melamine formaldehyde(MF) resin,Urea melamine formaldehyde resin(UMF) Phenolic resins - Phenol formaldehyde resin(PF) - resole and novolac Resorcinol formaldehyde(RF),Phenol resorcinol formaldehyde(PRF),Resin kettle Summarize the manufacture and industrial applications of Thermoplastic

Explain the working of Resin kettle 3 Understanding Synthetic resins - Introduction, chemistry, method of manufacture, lab preparation, properties& industrial applications of: Thermosetting resins Amino resins - Urea formaldehyde (UF) and Melamine formaldehyde(MF) resin,Urea melamine formaldehyde resin(UMF) Phenolic resins - Phenol formaldehyde resin(PF) - resole and novolac Resorcinol formaldehyde(RF),Phenol resorcinol formaldehyde(PRF),Resin kettle Summarize the manufacture and industrial applications of Thermoplastic is an official topic in Module 3 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the working of Resin kettle 3 Understanding Synthetic resins - Introduction, chemistry, method of manufacture, lab preparation, properties& industrial applications of: Thermosetting resins Amino resins - Urea formaldehyde (UF) and Melamine formaldehyde(MF) resin,Urea melamine formaldehyde resin(UMF) Phenolic resins - Phenol formaldehyde resin(PF) - resole and novolac Resorcinol formaldehyde(RF),Phenol resorcinol formaldehyde(PRF),Resin kettle Summarize the manufacture and industrial applications of Thermoplastic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the working of resin kettle 3 understanding synthetic resins - introduction, chemistry, method of manufacture, lab preparation, properties& industrial applications of: thermosetting resins amino resins - urea formaldehyde (uf) and melamine formaldehyde(mf) resin,urea melamine formaldehyde resin(umf) phenolic resins - phenol formaldehyde resin(pf) - resole and novolac resorcinol formaldehyde(rf),phenol resorcinol formaldehyde(prf),resin kettle summarize the manufacture and industrial applications of thermoplastic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the working of Resin kettle 3 Understanding Synthetic resins - Introduction, chemistry, method of manufacture, lab preparation, properties& industrial applications of: Thermosetting resins Amino resins - Urea formaldehyde (UF) and Melamine formaldehyde(MF) resin,Urea melamine formaldehyde resin(UMF) Phenolic resins - Phenol formaldehyde resin(PF) - resole and novolac Resorcinol formaldehyde(RF),Phenol resorcinol formaldehyde(PRF),Resin kettle Summarize the manufacture and industrial applications of Thermoplastic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain the working of Resin kettle 3 Understanding Synthetic resins - Introduction, chemistry, method of manufacture, lab preparation, properties& industrial applications of: Thermosetting resins Amino resins - Urea formaldehyde (UF) and Melamine formaldehyde(MF) resin,Urea melamine formaldehyde resin(UMF) Phenolic resins - Phenol formaldehyde resin(PF) - resole and novolac Resorcinol formaldehyde(RF),Phenol resorcinol formaldehyde(PRF),Resin kettle Summarize the manufacture and industrial applications of Thermoplastic □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
resins
  M3.01      Classify synthetic resins and its types                       2
Understanding
             Summarize the manufacture of thermosetting
  M3.02                                                                    8
Understanding
             resins
             Discuss the applications of thermosetting resins
  M3.03                                                                    4
Applying
             in wood industry
  M3.04      Explain the working of Resin kettle                           3
Understanding

Contents:
Synthetic resins - Introduction, chemistry, method of           manufacture, lab
preparation,
properties& industrial applications of: Thermosetting resins
Amino resins - Urea formaldehyde (UF) and Melamine formaldehyde(MF) resin,Urea
melamine formaldehyde resin(UMF)
Phenolic resins - Phenol formaldehyde resin(PF) - resole and novolac
Resorcinol formaldehyde(RF),Phenol resorcinol formaldehyde(PRF),Resin kettle

             Summarize the manufacture and industrial applications of Thermoplastic
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

resins

This module is presented from the official syllabus scope for Adhesive Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified **CO-linked**

official hours | see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: resins
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

3 Understanding thermoplastic resins Compare the advantages of each adhesive over

3 Understanding thermoplastic resins Compare the advantages of each adhesive over is an official topic in Module 4 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding thermoplastic resins Compare the advantages of each adhesive over using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding thermoplastic resins compare the advantages of each adhesive over should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding thermoplastic resins Compare the advantages of each adhesive over means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding thermoplastic resins Compare the advantages of each adhesive over □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding the other Explain the properties and applications of

3 Understanding the other Explain the properties and applications of is an official topic in Module 4 of Adhesive Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding the other Explain the properties and applications of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding the other explain the properties and applications of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding the other Explain the properties and applications of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding the other Explain the properties and applications of □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

3 Understanding thermoplastic resins CNSL based adhesives, Epoxy resin, Polyurethane adhesives, Advantage of polyurethane adhesives over urea and phenolic resins. Thermoplastic resins - Introduction, Polymerization of cellulose adhesives, polyethylene, polyamides, polyesters and vinyl acetate - properties and industrial applications Text / Reference: T/R Book Title/Author T1 F.P Kollmann, Principles of wood science & Technology ,Springer Adhesives for the Fbrest Products Industry, Borden, Inc., Chemical Division, R1 Adhesive & Chemical Products, New York, N.Y. Booth, C. C., Adhesives Available For Wood Bonding, Borden, Inc. Chemical R2 Division Adhesive & Chemical Products, New York, N.Y. Synthetic resins & their industrial applications – SBP – Board of consultants & R3 Engineers New Delhi R4 Lecture notes on plywood technology – IPIRI Bangalore1993 Carroll, W. K., How To Apply Adhesives, Product Engineering, McGraw-Bill R5 Inc., Rept., Nov. 22, 1965. Clark, L. E., Technical Director, Perkins Glue Co., A Practical Wood Adhesive R6 Guide, Furniture Desiv, & Manufacturing Magazine, January 1966. Online Resources: Sl.No Website Link 1 https://files.eric.ed.gov/fulltext/ED031557.pdf https://polymerinnovationblog.com/wp-content/uploads/2015/02/handbook-of-2 adhesive-technology.pdf https://www.adhesives.org/adhesives-sealants/adhesives-sealants- 3 overview/adhesive-technologies

3 Understanding thermoplastic resins CNSL based adhesives, Epoxy resin, Polyurethane adhesives, Advantage of polyurethane adhesives over urea and phenolic resins. Thermoplastic resins - Introduction, Polymerization of cellulose adhesives, polyethylene, polyamides, polyesters and vinyl acetate -properties and industrial applications Text / Reference: T/R Book Title/Author T1 F.P Kollmann, Principles of wood science & Technology ,Springer Adhesives for the Fbrest Products Industry, Borden, Inc., Chemical Division, R1 Adhesive & Chemical Products, New York, N.Y. Booth, C. C., Adhesives Available For Wood Bonding, Borden, Inc. Chemical R2 Division Adhesive & Chemical Products, New York, N.Y. Synthetic resins & their industrial applications – SBP – Board of consultants & R3 Engineers New Delhi R4 Lecture notes on plywood technology – IPIRI Bangalore1993 Carroll, W. K., How To Apply Adhesives, Product Engineering, McGraw-Bill R5 Inc., Rept., Nov. 22, 1965. Clark, L. E., Technical Director, Perkins Glue Co., A Practical Wood Adhesive R6 Guide, Furniture Desiv, & Manufacturing Magazine, January 1966. Online Resources: Sl.No Website Link 1 https://files.eric.ed.gov/fulltext/ED031557.pdf https://polymerinnovationblog.com/wp-content/uploads/2015/02/handbook-of- 2 adhesive-technology.pdf https://www.adhesives.org/adhesives-sealants/adhesives-sealants- 3 overview/adhesive-technologies is an official topic in Module 4 of Adhesive Technology. Study the terminology first, then connect the

stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding thermoplastic resins CNSL based adhesives, Epoxy resin, Polyurethane adhesives, Advantage of polyurethane adhesives over urea and phenolic resins. Thermoplastic resins - Introduction, Polymerization of cellulose adhesives, polyethylene, polyamides, polyesters and vinyl acetate - properties and industrial applications Text / Reference: T/R Book Title/Author T1 F.P Kollmann, Principles of wood science & Technology ,Springer Adhesives for the Fbrest Products Industry, Borden, Inc., Chemical Division, R1 Adhesive & Chemical Products, New York, N.Y. Booth, C. C., Adhesives Available For Wood Bonding, Borden, Inc. Chemical R2 Division Adhesive & Chemical Products, New York, N.Y. Synthetic resins & their industrial applications – SBP – Board of consultants & R3 Engineers New Delhi R4 Lecture notes on plywood technology – IPIRI Bangalore1993 Carroll, W. K., How To Apply Adhesives, Product Engineering, McGraw-Bill R5 Inc., Rept., Nov. 22, 1965. Clark, L. E., Technical Director, Perkins Glue Co., A Practical Wood Adhesive R6 Guide, Furniture Desiv, & Manufacturing Magazine, January 1966. Online Resources: Sl.No Website Link 1 https://files.eric.ed.gov/fulltext/ED031557.pdf https://polymerinnovationblog.com/wp-content/uploads/2015/02/handbook-of- 2 adhesive-technology.pdf https://www.adhesives.org/adhesives-sealants/adhesives-sealants- 3 overview/adhesive-technologies using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding thermoplastic resins cnsl based adhesives, epoxy resin, polyurethane adhesives, advantage of polyurethane adhesives over urea and phenolic resins. thermoplastic resins - introduction, polymerization of cellulose adhesives, polyethylene, polyamides, polyesters and vinyl acetate -properties and industrial applications text / reference: t/r book title/author t1 f.p kollmann, principles of wood science & technology ,springer adhesives for the fbrest products industry, borden, inc., chemical division, r1 adhesive & chemical products, new york, n.y. booth, c. c., adhesives available for wood bonding, borden, inc. chemical r2 division adhesive & chemical products, new york, n.y. synthetic resins & their industrial applications – sbp – board of consultants & r3 engineers new delhi r4 lecture notes on plywood technology – ipiri bangalore1993 carroll, w. k., how to apply adhesives, product engineering, mcgraw-bill r5 inc., rept., nov. 22, 1965. clark, l. e., technical director, perkins glue co., a practical wood adhesive r6 guide, furniture desiv, & manufacturing magazine, january 1966. online resources: sl.no website link 1 https://files.eric.ed.gov/fulltext/ed031557.pdf https://polymerinnovationblog.com/wp-content/uploads/2015/02/handbook-of- 2 adhesive-technology.pdf https://www.adhesives.org/adhesives-sealants/adhesives-sealants- 3 overview/adhesive-technologies should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding thermoplastic resins CNSL based adhesives, Epoxy resin, Polyurethane adhesives, Advantage of polyurethane adhesives over urea and phenolic resins. Thermoplastic resins - Introduction, Polymerization of cellulose adhesives, polyethylene, polyamides, polyesters and vinyl acetate -properties and industrial applications Text / Reference: T/R Book Title/Author T1 F.P Kollmann, Principles of wood science & Technology ,Springer Adhesives for the Fbrest Products Industry, Borden, Inc., Chemical Division, R1 Adhesive & Chemical Products, New York, N.Y. Booth, C. C., Adhesives Available For Wood Bonding, Borden, Inc. Chemical R2 Division Adhesive & Chemical Products, New York, N.Y. Synthetic resins & their industrial applications – SBP – Board of consultants & R3 Engineers New Delhi R4 Lecture notes on plywood technology – IPIRI Bangalore1993 Carroll, W. K., How To Apply Adhesives, Product Engineering, McGraw-Bill R5 Inc., Rept., Nov. 22, 1965. Clark, L. E., Technical Director, Perkins Glue Co., A Practical Wood Adhesive R6 Guide, Furniture Desiv, & Manufacturing Magazine, January 1966. Online Resources: Sl.No Website Link 1 https://files.eric.ed.gov/fulltext/ED031557.pdf https://polymerinnovationblog.com/wp-content/uploads/2015/02/handbook-of- 2 adhesive-technology.pdf https://www.adhesives.org/adhesives-sealants/adhesives-sealants- 3 overview/adhesive-technologies means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding thermoplastic resins CNSL based adhesives, Epoxy resin, Polyurethane adhesives, Advantage of polyurethane adhesives over urea and phenolic resins. Thermoplastic resins - Introduction, Polymerization of cellulose adhesives, polyethylene, polyamides, polyesters and vinyl acetate -properties and industrial applications Text / Reference: T/R Book

Title/Author T1 F.P Kollmann, Principles of wood science & Technology ,Springer Adhesives for the Fbrest Products Industry, Borden, Inc., Chemical Division, R1 Adhesive & Chemical Products, New York, N.Y. Booth, C. C., Adhesives Available For Wood Bonding, Borden, Inc. Chemical R2 Division Adhesive & Chemical Products, New York, N.Y. Synthetic resins & their industrial applications – SBP – Board of consultants & R3 Engineers New Delhi R4 Lecture notes on plywood technology – IPIRI Bangalore1993 Carroll, W. K., How To Apply Adhesives, Product Engineering, McGraw-Bill R5 Inc., Rept., Nov. 22, 1965. Clark, L. E., Technical Director, Perkins Glue Co., A Practical Wood Adhesive R6 Guide, Furniture Desiv, & Manufacturing Magazine, January 1966. Online Resources: Sl.No Website Link 1 https://files.eric.ed.gov/fulltext/ED031557.pdf https://polymerinnovationblog.com/wp-content/uploads/2015/02/handbook-of- 2 adhesive-technology.pdf https://www.adhesives.org/adhesives-sealants/adhesives-sealants- 3 overview/adhesive-technologies □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
resins

             Explain the manufacture of epoxy resins and
  M4.01                                                                3
Understanding
             thermoplastic resins
             Compare the advantages of each adhesive over
  M4.02                                                                3
Understanding
             the other
             Explain the properties and applications of
   M4.03                                                              3
Understanding
             thermoplastic resins
             Series Test – II                                         1

 Contents:
 CNSL based adhesives, Epoxy resin, Polyurethane adhesives, Advantage of
polyurethane
 adhesives over urea and phenolic resins.
 Thermoplastic resins - Introduction, Polymerization of cellulose adhesives,
polyethylene,
 polyamides, polyesters and vinyl acetate -properties and industrial applications

Text / Reference:

  T/R                                      Book Title/Author
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 1 industry Understanding. (3 marks)

Answer: Begin with the standard definition or identity of *1 industry Understanding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Classify adhesives 6 Understanding Explain the composition of adhesives and additives. (3 marks)

Answer: Begin with the standard definition or identity of *Classify adhesives 6 Understanding Explain the composition of adhesives and additives*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Applying used. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying used*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Discuss the concepts of glue mixing 4 Applying Adhesive types and their composition Adhesive industries in India - an overview Adhesion, adhesives, adherend, glue line; Classification of adhesives - Natural- starches, dextrins, vegetable gums, protein glues- animal glues, blood albumin, casein, vegetable protein glue, shellac rubber Asphalt sodium silicate, magnesium oxychloride; Synthetic – Thermosetting (urea, melamine,phenol, resorcinol..) and Thermoplastic resin(cellulose esters and ethers, polyvinyl alcohol..) Composition of adhesive - back bone polymer, solvent, thinner, catalyst, hardner, extender, filler, plasticiser, fortifier, preservatives; Glue mixing- types of mixers - process by which adhesive develop strength, addition of caustic soda or acid to control the PH during mixing. Viscosity, shelf life and pot life of resin.. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss the concepts of glue mixing 4 Applying Adhesive types and their composition Adhesive industries in India - an overview Adhesion, adhesives, adherend, glue line; Classification of adhesives - Natural- starches, dextrins, vegetable gums, protein glues- animal glues, blood albumin, casein, vegetable protein glue, shellac rubber Asphalt sodium silicate, magnesium oxychloride; Synthetic – Thermosetting (urea, melamine,phenol, resorcinol..) and Thermoplastic resin(cellulose esters and ethers, polyvinyl alcohol..) Composition of adhesive - back bone polymer, solvent, thinner, catalyst, hardner, extender, filler, plasticiser, fortifier, preservatives; Glue mixing- types of mixers - process by which adhesive develop strength, addition of caustic soda or acid to control the PH during mixing. Viscosity, shelf life and pot life of resin.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 2

Define or explain Explain the rheology of glues. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the rheology of glues*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Discuss the glue spreading 5 Understanding Demonstrate the tests on glues and gluing. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss the glue spreading 5 Understanding Demonstrate the tests on glues and gluing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Applying defects Physical chemical principles of gluing Cohesion - adhesion, Rheology, intermolecular forces, polarity, primary and secondary forces (valences), phenomenon of hardening Influence of - temperature& concentration on surface tension; PH and amount of polymerization on hardening of glue joints, physical properties of adhesive film- thickness of adhesive film Pre-treatments of wood prior to gluing. Glue spreading, Defects in gluing - effect of wood species and moisture content on the strength of glue joint - strength improvement and stresses in glue joints. Tests on adhesive - PH, Viscosity, water tolerance, solid content by oven dry method, specific gravity, solubility, estimation of phenol and formaldehyde. Describe the manufacture and industrial applications of Thermosetting. (3 marks)

Answer: Begin with the standard definition or identity of *6 Applying defects Physical chemical principles of gluing Cohesion - adhesion, Rheology, intermolecular forces, polarity, primary and secondary forces (valences), phenomenon of hardening Influence of - temperature& concentration on surface tension; PH and amount of polymerization on hardening of glue joints, physical properties of adhesive film- thickness of adhesive film Pre-treatments of wood prior to gluing. Glue spreading, Defects in gluing - effect of wood species and moisture content on the strength of glue joint - strength improvement and stresses in glue joints. Tests on adhesive - PH, Viscosity, water tolerance, solid content by oven dry method, specific gravity, solubility, estimation of phenol and formaldehyde. Describe the manufacture and industrial applications of Thermosetting*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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Module 3

Define or explain Classify synthetic resins and its types 2 Understanding Summarize the manufacture of thermosetting. (3 marks)

Answer: Begin with the standard definition or identity of *Classify synthetic resins and its types 2 Understanding Summarize the manufacture of thermosetting*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□□ □□□□□□□.

Define or explain 8 Understanding resins Discuss the applications of thermosetting resins. (3 marks)

Answer: Begin with the standard definition or identity of *8 Understanding resins Discuss the applications of thermosetting resins*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□□ □□□□□□□.

Define or explain 4 Applying in wood industry. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying in wood industry*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□□ □□□□□□□.

Define or explain Explain the working of Resin kettle 3 Understanding Synthetic resins - Introduction, chemistry, method of manufacture, lab preparation, properties& industrial applications of: Thermosetting resins Amino resins - Urea formaldehyde (UF) and Melamine formaldehyde(MF) resin,Urea melamine formaldehyde resin(UMF) Phenolic resins - Phenol formaldehyde resin(PF) - resole and novolac Resorcinol formaldehyde(RF),Phenol resorcinol formaldehyde(PRF),Resin kettle

Summarize the manufacture and industrial applications of Thermoplastic. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the working of Resin kettle 3 Understanding Synthetic resins - Introduction, chemistry, method of manufacture, lab preparation, properties& industrial applications of: Thermosetting resins Amino resins - Urea formaldehyde (UF) and Melamine formaldehyde(MF) resin,Urea melamine formaldehyde resin(UMF) Phenolic resins - Phenol formaldehyde resin(PF) - resole and novolac Resorcinol formaldehyde(RF),Phenol resorcinol formaldehyde(PRF),Resin kettle Summarize the manufacture and industrial applications of Thermoplastic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 3 Understanding thermoplastic resins Compare the advantages of each adhesive over. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding thermoplastic resins Compare the advantages of each adhesive over*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding the other Explain the properties and applications of. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding the other Explain the properties and applications of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding thermoplastic resins CNSL based adhesives, Epoxy resin, Polyurethane adhesives, Advantage of polyurethane adhesives over urea and phenolic resins. Thermoplastic resins - Introduction, Polymerization of cellulose adhesives, polyethylene, polyamides, polyesters and vinyl acetate -properties and industrial applications Text / Reference: T/R Book Title/Author T1 F.P Kollmann, Principles of wood science & Technology ,Springer Adhesives for the Fbrest Products Industry, Borden, Inc., Chemical Division, R1 Adhesive & Chemical Products, New York, N.Y. Booth, C. C., Adhesives Available For Wood Bonding, Borden, Inc. Chemical R2 Division Adhesive & Chemical Products, New York, N.Y. Synthetic resins & their industrial applications – SBP – Board of consultants & R3 Engineers New Delhi R4 Lecture notes on plywood technology – IPIRI Bangalore1993 Carroll, W. K., How To Apply Adhesives, Product Engineering, McGraw-Bill R5 Inc., Rept., Nov. 22, 1965. Clark, L. E., Technical Director, Perkins Glue Co., A Practical Wood Adhesive R6 Guide, Furniture Desiv, & Manufacturing Magazine, January 1966. Online Resources: Sl.No Website Link 1 https://files.eric.ed.gov/fulltext/ED031557.pdf https://polymerinnovationblog.com/wp-content/uploads/2015/02/handbook-of-2 adhesive-technology.pdf https://www.adhesives.org/adhesives-sealants/adhesives-sealants- 3 overview/adhesive-technologies. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding thermoplastic resins CNSL based adhesives, Epoxy resin, Polyurethane adhesives, Advantage of polyurethane adhesives over urea and phenolic resins. Thermoplastic resins - Introduction, Polymerization of cellulose adhesives, polyethylene, polyamides, polyesters and vinyl acetate -properties and industrial applications Text / Reference: T/R Book Title/Author T1 F.P Kollmann, Principles of wood science & Technology ,Springer Adhesives for the Fbrest Products Industry, Borden, Inc., Chemical Division, R1 Adhesive & Chemical Products, New York, N.Y. Booth, C. C., Adhesives Available For Wood Bonding, Borden, Inc. Chemical R2 Division Adhesive & Chemical Products, New York, N.Y. Synthetic resins & their industrial applications – SBP – Board of consultants & R3 Engineers New Delhi R4 Lecture notes on plywood technology – IPIRI Bangalore1993 Carroll, W. K., How To Apply Adhesives, Product Engineering, McGraw-Bill R5 Inc., Rept., Nov. 22, 1965. Clark, L. E., Technical Director, Perkins Glue Co., A Practical Wood Adhesive R6 Guide, Furniture Desiv, & Manufacturing Magazine, January 1966. Online Resources: Sl.No Website Link 1 https://files.eric.ed.gov/fulltext/ED031557.pdf https://polymerinnovationblog.com/wp-content/uploads/2015/02/handbook-of- 2 adhesive-technology.pdf https://www.adhesives.org/adhesives-sealants/adhesives-*

sealants- 3 overview/adhesive-technologies. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 industry Understanding**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain the rheology of glues**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Classify synthetic resins and its types 2 Understanding Summarize the manufacture of thermosetting**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **3 Understanding thermoplastic resins Compare the advantages of each adhesive over**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://files.eric.ed.gov/fulltext/ED031557.pdf
- https://polymerinnovationblog.com/wp-content/uploads/2015/02/handbook-of-adhesive-technology.pdf
- https://www.adhesives.org/adhesives-sealants/adhesives-sealants- overview/adhesive-technologies

IN-PAGE SEARCH

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